



Java Fundamentals: Basic Programming Assignment - 1

Problem Statement:

As a Java Developer at XYZ Solutions, you have been assigned to develop the following programs given below. You have to make sure that the programs are efficient in terms of Space Complexity as well as Time Complexity.

Requirements:

1. Write a Java program to print "Hello, World!" to the console.

Examples:

Output: Hello, World!

2. **Program to find the sum of two numbers:** The program should prompt the user to enter two numerical values and calculate their sum.

Requirements:

- a. The program should prompt the user to enter the first number.
- b. Take the input for the first number from the user.
- c. The program should prompt the user to enter the second number.
- d. Take the input for the second number from the user.
- e. Calculate the sum of the two numbers.
- f. Display the sum as the output.

Examples:

Input: 5, 7

Output: Sum: 12

Input: -10, 20

Output: Sum: 10

Constraints:

Integers within the range of -2^{31} to $2^{31} - 1$

3. **Implement a Java program to swap the values of two variables without using a temporary variable.**

Examples:

Input: a = 5, b = 10

Output: After swapping: a = 10, b = 5

Input: a = -7, b = 3

Output: After swapping: a = 3, b = -7

Constraints: None

4. Write a Java program to find the maximum of three numbers.

Examples:

Input: 5, 12, 8

Output: Maximum: 12

Input: -3, -10, -6

Output: Maximum: -3

Constraints: Integers within the range of -2^{31} to $2^{31} - 1$

5. Program to check if a number is prime: A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself.

Requirements:

- The program should prompt the user to enter an integer greater than 1.
- Check whether the entered number is a prime number or not.
- Display an appropriate message if the number is prime or not.

Examples:

Input: 7

Output: Prime

Input: 10

Output: Not Prime

Constraints: The input number is a positive integer greater than 1.

6. Create a program that generates and prints the first N terms of the Fibonacci sequence.

The Fibonacci series is a sequence of numbers where each number is the sum of the two preceding ones.

Requirements:

- The program should prompt the user to enter the number of terms they want in the Fibonacci series.
- Generate the Fibonacci series up to the specified number of terms using a loop.
- Display the generated Fibonacci series as the output.

Examples:**Input:** N = 7**Output:** Fibonacci sequence: 0, 1, 1, 2, 3, 5, 8**Input:** N = 10**Output:** Fibonacci sequence: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34**Constraints:** N is a positive integer.

7. **Program to find the factorial of a number:** The factorial of a non-negative integer 'n' is the product of all positive integers less than or equal to 'n'.

Requirements:

- The program should prompt the user to enter a non-negative integer 'n'.
- Calculate the factorial of 'n' using an iterative approach.
- Display the factorial of 'n' as the output.
- Handle the case when the user enters a negative number or zero.

Examples:**Input:** 5**Output:** Factorial: 120**Input:** 0**Output:** Factorial: 1**Constraints:** The input is a non-negative integer.

8. **Implement a program to find the sum of even numbers between 1 and N.**

Examples:**Input:** N = 10**Output:** Sum of even numbers: 30

Input: N = 15

Output: Sum of even numbers: 56

Constraints: N is a positive integer.

9. Write a Java program to reverse a given string.

Examples:

Input: "hello"

Output: Reversed string: "olleh"

Input: "Intellipaat"

Output: Reversed string: "taapilletnI"

Constraints: None

10. Create a program that checks if a given string is a palindrome or not.

Examples:

Input: "radar"

Output: Palindrome

Input: "Intellipaat"

Output: Not Palindrome

Constraints: None